TECHNICAL APPENDIX

Mobility Shift — Dubai RTA Bus Network Analytics

1. Dataset & Model Specifications

1.1 Dataset Overview

Five source files totalling over 209,534 rows were used across all three stages. All files were joined using composite key: Route_ID + Stop_ID + Date.

Stage 1 — Historical Datasets	
Train_Ridership_2022_to_H1_2025.csv	195,381 rows | Primary fact table: Boarding_Count + Alighting_Count per stop per date (2022–H1 2025)
Train_Traffic_2022_to_H1_2025.csv	1,099 rows | Congestion_Level (1–5) and Avg_Speed_kmph per daily date
Bus_Routes.csv	12 rows | Route_ID, Route_Type, Route_Code, Route_Length_km, Avg_Travel_Time_Min
Bus_Stops.csv	60 rows | Stop_ID, Zone, Stop_Type, Lat, Long
Route_Stop_Mapping.csv	junction table | Route_ID + Stop_ID + Stop_Sequence + Dwell_Time_Min

Stage 2 — Shock Datasets	
Shock_Ridership_2025_Q3.csv	14,076 rows | Jul–Sep 2025 ridership | New columns: Route_Code, Route_Type (direct)
Shock_Traffic_2025_Q3.csv	92 rows | Q3 daily congestion + speed | Avg congestion 3.09 | Avg speed 29.16 km/h

Stage 3 — Out-of-Time Validation Datasets	
OutOfTime_Ridership_2025_Q4.csv	14,076 rows | Oct–Dec 2025 actual ridership | Withheld during Stage 2
OutOfTime_Traffic_2025_Q4.csv	92 rows | Q4 daily traffic | Avg congestion 2.83 | Avg speed 30.99 km/h

1.2 Derived Metrics & Feature Engineering

All derived features were computed in Python (pandas) before export to Power BI and Excel.

Feature	Formula / Logic	Used In
Total_Pax	Boarding_Count + Alighting_Count	All stages — primary demand signal
CAGR (2022–2024)	(Pax_2024 / Pax_2022)^0.5 − 1	Stage 1: route growth comparison
Seasonal Index	Monthly Avg Pax ÷ Overall Annual Avg Pax	Stage 1 & 2: seasonal adjustment
Load Ratio	Route Avg Daily Pax ÷ Capacity Benchmark	Stage 1: risk classification

YoY Growth %	(Current Year Pax − Prior Year Pax) ÷ Prior Year	Stage 1 & 2: growth tracking
Congestion Elasticity	Pearson correlation(Total_Pax, Congestion_Level)	Stage 1: modal shift analysis
Net Flow	Boarding_Count − Alighting_Count per zone/date	Stage 1: corridor imbalance
Dev % (Stage 2 Bias)	(Q3 Actual − Stage 1 Forecast) ÷ Stage 1 Forecast	Stage 2: structural break detection
MAPE (Stage 3)	|Actual_Q4 − Stage2_Forecast| ÷ Stage2_Forecast × 100	Stage 3: model accuracy audit
Directional Bias	(Actual_Q4 − Stage2_Forecast) ÷ Stage2_Forecast × 100	Stage 3: under/over-reaction classification

2. Methodology

2.1 Stage 1 — Baseline Analytical Approach

Stage 1 used a five-module analytical framework, each directly addressing one of the five mandatory business deliverables:

1. **Module 1 — Growth Decomposition:** Yearly total Pax aggregated, YoY growth rate computed. Route-level CAGR calculated as geometric mean growth 2022→2024. All 12 routes showed CAGR between 10.7% and 11.4%, confirming uniform growth pre-Metro.
2. **Module 2 — Seasonality Analysis:** Seasonal Index computed as monthly average Pax ÷ overall annual average. Weekday/Weekend divergence measured by route type. Jan = 1.158 peak; Jul = 0.834 trough.
3. **Module 3 — Corridor Analysis:** Zone-level Pax aggregated. Net Flow = Boarding − Alighting per zone to identify origin vs. destination corridors. CBD_Downtown confirmed as 29% of total network Pax.
4. **Module 4 — Congestion Elasticity:** Pearson correlation between Total_Pax and Congestion_Level computed at network level (+0.137) and by route type. Level 5 congestion produced 20.4% more Pax than Level 1, confirming positive modal shift elasticity.
5. **Module 5 — H2 2025 Forecast:** Base forecast derived using H2 2024 actuals × observed YoY growth rate × monthly seasonal index. Formula: H2_2025_Forecast = H2_2024 × (1 + Growth_23→24) × Seasonal_Index.

2.2 Stage 2 — Structural Break Detection Method

Stage 2 employed a four-type structural shift classification framework:

Shift Type	Detection Method	Threshold Applied
Level Shift	Compare Q3 2025 daily avg vs H1 2025 daily avg by route code	Deviation > ±10% triggers structural flag
Volatility Shift	Coefficient of Variation (CoV = StdDev ÷ Mean) by route type before/after	CoV increase > 20% = elevated volatility
Elasticity Shift	Re-compute congestion correlation on Q3 2025 data vs historical	If |Δcorr| > 0.10 = elasticity shift detected
Congestion-Mediated Shift	Monthly congestion progression Jul→Sep vs H1 baseline	Level increase > 0.5 on 3-month basis = congestion shift

Verdict classification: A route was classified as REGIME CHANGE if deviation exceeded ±10% AND September Q3 showed no reversion toward baseline. A route was classified as TEMPORARY if deviation was below ±10% or if September showed partial recovery.

2.3 Stage 2 — Forecast Recalibration Method

Post-shock forecast used the following decision matrix per route type:

Route Type	Base Used	Seasonal Factor	Justification
City (C01–C04)	Stage 1 H1 2025 avg	Stage 1 seasonal index (Oct 0.960, Nov 1.072, Dec 1.133)	Only +2.2% deviation; Metro unaffected; Stage 1 model valid
Express (X11, X28, X66)	Q3 2025 daily avg	Stage 1 seasonal index applied	REGIME CHANGE confirmed; +29.8% permanent Metro feeder shift
Feeder (F12, F18, F25)	Q3 2025 daily avg	Stage 1 seasonal index applied	REGIME CHANGE confirmed; −23.5% structural CBD loss to Metro
Intercity (E16, E22)	Stage 1 × 0.80 blended	Stage 1 seasonal index applied	Mixed: seasonal + Metro; partial Q4 recovery expected

3. Assumptions

3.1 Core Modelling Assumptions

Assumption	Basis	Risk if Wrong
Seasonal index from 2022–2024 remains valid post-Metro	3-year historical index stable across all route types in Stage 1	Low risk for City/Intercity; Medium for Express/Feeder (new demand patterns)
Express surge is permanent regime (not temporary Metro novelty)	Sep 2025 higher than Jul 2025 — no reversion in any month of Q3	If Metro novelty decays in Q4, Express overforecast by ~15–20%
Feeder CBD contraction is permanent	F25 showed no recovery in Sep; structural cause (Metro direct competition) is permanent	If F25 partially recovers, fleet reallocation may over-free Feeder capacity
Intercity -8.2% is 80% temporary (seasonal) + 20% Metro	Conservative blended approach; validated against Q4 actuals	Q4 actuals showed +11.2–13.3% recovery — assumption directionally correct
Load ratio threshold: >1.2 = OVERLOAD, <0.75 = UNDERUTILISED	Standard bus network capacity planning benchmarks	Threshold sensitivity: ±0.05 shift does not materially change priority routing
Q3/Q4 ratio of 1.207 approximates seasonal uplift	Derived from 2024 H2/H1 ratio applied to post-shock base	If 2025 winter is stronger, underforecasts risk for Express routes in December

3.2 Data Quality Assumptions

1. Total_ Pax = Boarding + Alighting correctly represents double-counted passenger touchpoints (each trip appears twice). Consistent with case methodology guidelines.
2. Q3 Shock data (14,076 records) represents a complete 92-day quarter with no gaps. Date range confirmed: July 1 – September 30, 2025.
3. Q4 Out-of-Time data (14,076 records) represents a complete 92-day quarter (Oct 1 – Dec 31, 2025). Used only for Stage 3 validation.
4. Bus_Routes.csv Route_Code mapping (C = City, X = Express, F = Feeder, E = Intercity) is consistent across all datasets. Cross-validated by Route_ ID join.

4. Validation Approach

4.1 Stage 1 Model Validation :

Stage 1 forecast was validated directionally using internal cross-validation: H1 2025 actual vs. the 2024→2025 growth projection derived from 2023→2024 trends. All 12 routes showed forecast deviation within ±5% for H1 2025, confirming model suitability for a stable (pre-Metro) demand environment.

4.2 Stage 2 Out-of-Time Validation (Stage 3 Audit)

The primary validation methodology was an out-of-time (OOT) test: Stage 2 recalibrated forecasts were held blind against Q4 2025 actuals released at Stage 3.

Route Type	Stage 2 Q4 Forecast	Q4 Actual	MAPE	Directional Bias	Verdict
Express (avg)	3,264	3,229	1.1%	-1.1%	ACCURATE — Regime confirmed
City (avg)	3,067	3,499	14.0%	+14.0%	UNDERREACTION — Seasonal beat Q3 base
Feeder (avg)	1,941	2,119	9.3%	+9.3%	UNDERREACTION — Partial Metro recovery
Intercity (avg)	1,842	2,069	12.3%	+12.3%	UNDERREACTION — Seasonal uplift stronger than blended model
Network (avg)			**9.33%**		8 of 12 routes underreacted; 0 overreacted; 4 accurate

> *Key Validation Insight: Express routes (the regime change segment) achieved MAPE of 1.1% — confirming the Stage 2 recalibration was correct. City underreaction (MAPE 14%) is explained by seasonal uplift exceeding Q3 base — the model was conservative, not wrong in direction.*

4.3 Monthly Accuracy Breakdown :

Month	Express Error	City Error	Feeder Error	Intercity Error	Notes

October 2025	-2.7%	+10.8%	+6.0%	+9.4%	Express underforecast mild; City seasonal began early
November 2025	-5.5%	+9.3%	+1.7%	+7.9%	Express overforecast slightly; City peak confirmed
December 2025	-9.8%	+4.9%	+2.9%	+2.8%	Express demand stabilised; City/Feeder/Intercity converging to forecast

5. Sensitivity Analysis

5.1 Seasonal Index Sensitivity

The forecast is most sensitive to the seasonal index assumption for Express routes, as Q3 is the new base and any index error compounds on a higher absolute level. The table below shows December 2025 Express forecast under three index scenarios:

Scenario	Dec Seasonal Index	Express Dec Forecast	City Dec Forecast	Fleet Impact
Bear Case	1.05 (−7.8%)	3,388	3,216	1 bus below Plan — manageable
Base Case (used)	1.133 (central estimate)	**3,698**	3,475	Fleet plan at current recommendation
Bull Case	1.20 (+5.9%)	3,916	3,675	Requires +1 additional Express bus in Dec

5.2 Metro Adoption Sensitivity

The regime change classification is contingent on Metro adoption remaining at Q3 2025 levels. Two scenarios were modelled:

5. Scenario A — Metro adoption accelerates in Q4: Feeder demand drops further (F25 could fall to 1,800/day). Express demand stabilises. This was partially observed: Express MAPE −9.8% in December suggests adoption is growing. Action: Pre-stage Feeder reduction in Jan 2026.
6. Scenario B — Metro novelty decays in Q4: Express demand reverts toward Stage 1 (2,600/day). Feeder recovers toward pre-Metro levels. This was NOT observed in Q4 data — Express held at 3,229, confirming regime permanence.

5.3 Fleet Reallocation Sensitivity

The proposed reallocation transfers 4–6 buses from Feeder/Intercity to Express. The sensitivity is asymmetric:

7. If Express demand beats forecast (Bull Case): reallocation is still insufficient; 1 additional bus needed from City route weekend slack.

8. If Feeder demand recovers more than projected (as partially seen in Q4 +9.3%): partial reallocation from Feeder is less aggressive. Retain F12/F18 at current frequency; reduce F25 only.
9. If City seasonal beats Base Case by >15% (observed in Oct/Nov): activate Volatility Reserve buffer fleet immediately; monitor C01/C03 daily for load > 4,000.

6. Tools & Code Stack Declaration

Per competition AI & Tool Usage Policy, all major tools used are declared below:

Tool Stack Declaration	
Python (pandas, numpy)	Data loading, merging, feature engineering, structural break analysis, MAPE calculation, sensitivity modelling. All code in Jupyter Notebooks provided.
Plotly Express / Graph Objects	Interactive visualisations: Before/After route comparison bars, congestion trend lines, accountability audit charts, monthly progression charts.
Power BI	5-page interactive dashboard: Growth, Seasonality, Corridor, Congestion-Demand, Forecast & Fleet. DAX measures for all KPIs.
Excel	Q4 seasonal index model: =Q3_Base × Seasonal_Idx. Risk flag formula: IF(Dec>3500,"CRITICAL",IF(Dec>3000,"OVERLOAD","MANAGEABLE")). Sensitivity tables.
AI Tools Used: Chatgpt , Gemini, Claude Ai , Multi ,etc	In depth Analysis of Data and visualization of Data through plotly , image generation for better understanding , DAX measures for Power Bi.